

PRACTICAL 5

1. Aim

To understand the concepts of Mathematical and Recursive Functions by implementing Factorial and Fibonacci analysis using SCILAB.

2. Prior Concepts

Students should be familiar with:

- Mathematical Functions
- Recursive Functions
- Factorial
- Fibonacci Series
- For Loop
- User Defined Functions in SCILAB
- Basic SCILAB Programming

3. New Concepts

After completing this practical, students will learn:

- Creating user-defined functions in SCILAB
- Implementing iterative algorithms
- Implementing recursive algorithms
- Calculating factorial using iteration and recursion
- Generating Fibonacci series using iteration
- Finding Fibonacci numbers using recursion
- Comparing iterative and recursive approaches

4. Procedure

- Open SCILAB.
- Create a new script file.
- Define the required mathematical and recursive functions.
- Execute the SCILAB program.
- Observe the generated outputs.
- Compare the iterative and recursive implementations.
- Analyze the growth of Factorial and Fibonacci functions.
- Save the program and note the observations.

5. Implementation with Scilab Code

Question 1: Find the Factorial of a Number using an Iterative Function.

Process:

1. Define a function to calculate the factorial using iteration.
2. Initialize the factorial value to 1.
3. Multiply the numbers from 1 to n using a for loop.
4. Display the factorial of the given number.

Question 2: Find the Factorial of a Number using Recursion.

Process:

1. Define a recursive factorial function.
2. Check the base condition.
3. Return 1 when n equals 0.
4. Otherwise call the function recursively.
5. Display the factorial.

Question 3: Generate Fibonacci Series using Iteration.

Process:

1. Initialize the first two Fibonacci numbers.
2. Display the first two numbers.
3. Generate the remaining numbers using a for loop.
4. Display the Fibonacci series.

Question 4: Find the nth Fibonacci Number using Recursion.

Process:

1. Define a recursive Fibonacci function.
2. Check the base conditions.
3. Compute the Fibonacci number recursively.
4. Display the nth Fibonacci number.

Question 5: Compare Factorial Growth with Fibonacci Growth.

Process:

1. Define recursive functions for Factorial and Fibonacci.
2. Generate values for different n.
3. Display the corresponding Factorial and Fibonacci values.
4. Compare their growth.

Question 6: Analyze Recursive Function Calls in Fibonacci Computation.

Process:

1. Initialize a counter variable.
2. Create a recursive Fibonacci function.
3. Count the number of recursive function calls.
4. Display the Fibonacci number and total function calls.

6. Result

Successfully implemented mathematical and recursive functions in SCILAB to calculate the Factorial of a number, generate the Fibonacci series, determine the nth Fibonacci number, compare the growth of Factorial and Fibonacci functions, and analyse recursive function calls.

7. Applications

- ✓ Mathematical Computations
- ✓ Algorithm Design and Analysis
- ✓ Dynamic Programming
- ✓ Artificial Intelligence
- ✓ Computer Graphics
- ✓ Data Structures
- ✓ Cryptography
- ✓ Population Growth Analysis
- ✓ Financial Forecasting
- ✓ Scientific Computing

8. Practice Questions

Find the Factorial of 8 using an iterative function.	Compare the Factorial and Fibonacci values for $n = 10$.
Find the Factorial of 10 using recursion.	Modify the recursive Fibonacci program to count the total number of recursive calls for $n = 8$.
Generate the first 15 Fibonacci numbers using iteration.	Compare the execution time of iterative and recursive Factorial functions.
Find the 12th Fibonacci number using recursion.	Compare the execution time of iterative and recursive Fibonacci functions.

9. Questions (Viva / Oral)

1. What is a mathematical function?
2. What is a recursive function?
3. Define Factorial.
4. Define Fibonacci Series.
5. What is the base condition in recursion?
6. What is the difference between iteration and recursion?
7. Which approach is more efficient for calculating Factorial?
8. Why is recursive Fibonacci slower than iterative Fibonacci?
9. What is the purpose of a user-defined function in SCILAB?
10. Give two real-life applications of the Fibonacci sequence.